# The ELIM production workstation: what we'd build and why

## 1. The whole picture

This build puts our video, network and control gear into the ProX AVPRO rolling workstation (the AVPRO). It has two 12U bays, two 4U tops and a work surface. Of its three lids, the flip-up lid holds the two monitors and the other two come off the case. The Behringer Wing sits beside it in the ProX BXWING hydraulic flip case (the Wing case), joined by five short cables. Every field cable and those five land on one labeled, color-coded panel on the back of the AVPRO.

Right now our rig deploys once or twice a week at different venues, and we set up and tear down under time pressure. The build also removes the need for extra tables. The Wing flips up in its own case, and the AVPRO brings its own work surface and pop-up lid tables. The design puts the physical job first. The system has to set up, run the event and tear down reliably before anything else matters.

| | |
|---|---|
| **~40 → ~15** loose connections become labeled cables | **~30 min** saved at setup, and again at teardown |

Both figures are estimates, and we'll measure them. I'll time setup and teardown before the cutover and again after it.

The order is about $4,600 ($4,641.84). That's an estimate. Most prices are estimated based on online pricing, and the cases and freight are placeholders until locked in with Matt. Almost all the gear inside is already owned, so it isn't in the total. A $100.00 connector plate for the Wing case comes on top, only if the Wing case can take one.

**Flip-up lid**

| | | |
|---|---|---|
| **Left screen**<br>every camera at once | **Right screen**<br>ProPresenter second screen | **Wireless camera receiver**<br>on top of the lid when open |

**Desk**

| | | |
|---|---|---|
| **Switcher**<br>on the desk | **Audio adapter**<br>beside it | **Two laptops**<br>ProPresenter + streaming |

**Bay A: the active system**

| left open |
|---|
| Cooling fan |
| Streaming encoder (Resi) |
| Recorder + audio switch |
| LED wall transcoder |
| Network switch + router |
| Cable management |
| Airflow + growth |
| Power strip (backed) |
| Battery backup (UPS) |

**Bay B: storage + charging**

| left open |
|---|
| Metered strip + charger strip |
| Left open |
| Drawer |
| Drawer |




The AVPRO product listing photo, with the seller's labels.

- Camera / SDI: Blue
- HDMI: White
- Dante: Green
- Production net: Yellow
- Internet: Red
- LED data: Purple
- XLR + AES50: Orange
- Power: Black

**Back of Bay A: the connection panel**

| | | | | |
|---|---|---|---|---|
| 8 camera inputs | Confidence out | LED wall x2 | Internet in | Network to Wing |
| Audio network to Wing | Talkback + in-ears | Power to Wing | | |

The layout is decided but can be adapted and changed. Colors are cable colors. The two removable lids have legs inside them, so each can pop up into an accessory table.

# 2. Walking the build

This section walks the build from the outside in. Each starts with what happens there at a service or event. Then there's a short card for each part: what it does, why it's there, what we passed on if build/ records it, what it costs, what we give up, and where it stands.

## a. The AVPRO station

The AVPRO is the production position. Streaming and presentation work is done at this desk under the two monitors on the flip-up lid, and audio mixing at the Wing case beside it. This creates a space for three stations at the production position.

### The case: ProX AVPRO rolling workstation

**What it does.** It holds everything but the Wing. It has two 12U bays, two 4U tops, a work surface and three lids, and it rolls. The flip-up lid holds the two monitors.

**Why it's there.** The build centers on one workstation and everything that connects to it. Our design principles also favor commercial cases and mounts over DIY where they work.

**Cost.** About $500, a placeholder until Matt's quote.

**What we give up.** Closed, it's large enough to drive freight cost. The side compartments aren't used in the design.

**Status.** The design is decided. The price is open until Matt's quote, and side-door storage gets checked on delivery.

### Two lid monitors: LG UltraGear 24GS60F-B

**What it does.** Both monitors are on the flip-up lid. The LEFT lid monitor shows the ATEM multiview, every camera at once, from day one. The RIGHT lid monitor is for the ProPresenter Mac, as its second screen now and ready for a later move to a Mac Mini.

**Why it's there.** The multiview can't depend on a laptop, so the left screen runs straight from the switcher. Together the two monitors replace the mobile monitor setup we use today.

**Why not another.** A 2000-nit tier was considered and dropped on cost (section 3).

**Cost.** About $320 with adapters, estimates. A brighter or used option would change this.

**What we give up.** At 300 nit matte, the LG is the minimum worth considering and probably isn't bright enough under a tent.

**Status.** The model is open on brightness (section 4), and lid fit gets checked on delivery.

### Wireless camera receiver: Hollyland RX

**What it does.** It receives the wireless Pocket 4K and sends SDI to ATEM IN 2. When the case is open it mounts on top of the flip-up lid. Until setup it stows in a drawer or with the cameras.

**Why it's there.** It can't live in the rack, because it won't get good signal there.

**Cost.** Already owned. The lid mount isn't costed.

**Status.** The placement is decided. The lid mount and its route to ATEM IN 2 are open (section 4).

### The two removable lids as pop-up tables

**What it does.** The AVPRO has three lids. The flip-up lid stays on the case and holds the two monitors. The other two lids come off the case, and each has legs inside it so it can pop up into an accessory table, for example for the camera charging area or other needs.

**Why it's there.** They give us extra surface at a venue without bringing tables.

**Cost.** Expected to come with the case. If not, we add them (not costed).

**Status.** Assumed: legs come with the removable lids. Open until Matt confirms.

## b. The desk

The ATEM switcher, the ProPresenter Mac and the streaming laptop run from the desk. The switcher's I/O cables drop through a 3in grommet into the Bay A rack space.

### Switcher: ATEM Television Studio Pro 4K

**What it does.** It switches all eight SDI inputs. Its multiview feeds the LEFT lid monitor, its AUX feeds the HyperDeck HD Mini recorder, and OUT 8 feeds the stream encoder. It sits over the 3in grommet, and its I/O drops into Bay A.

**Why it's there.** It's the switcher we run today.

**Cost.** Already owned. The retention tray isn't costed.

**Status.** The placement is decided. The retention tray is open (section 4).

### Audio adapter: Dante AVIO

**What it does.** It takes Dante from the Bay A satellite switch and feeds two XLR pigtails into ATEM audio channels 1 and 2. It runs on PoE from that switch.

**Why it's there.** The new 8-port PoE switch in Bay A powers it, so it needs no injector.

**Cost.** Already owned.

**Status.** Decided and confirmed PoE-powered. Reading its model label is open.

### Video interface: UltraStudio Monitor 3G

**What it does.** It takes ProPresenter from the ProPresenter Mac over USB-C and sends SDI to ATEM IN 8.

**Why it's there.** It puts ProPresenter on the switcher as a clean SDI input.

**Cost.** Already owned.

**Status.** Placement is open (section 4). The current plan has it velcro'd behind the lid monitors.

### The laptops and their power

**What it does.** The streaming laptop runs ATEM Software Control, the HyperDeck and Resi. The ProPresenter Mac runs ProPresenter and plays music and media in house over Dante. It sends HDMI to the LED wall transcoder and uses an Anker hub for production Ethernet and the confidence monitor. Both laptops plug into the battery-backed CPS1215RM strip.

**Why it's there.** These are the laptops we use now. On the battery-backed strip they keep running if venue power drops.

**Cost.** Already owned.

**Status.** Decided.

## c. Bay A, the active system

Nobody touches Bay A for setup. It's wired once and left alone, and every network jumper stays inside it. All connection work is done from the rear panel instead.

### Battery backup and backed strip: CyberPower CP1500PFCRM2U + CPS1215RM

**What it does.** The 2U pure-sine UPS sits at the bottom of Bay A, fed from power drop 1. The basic 1U CPS1215RM strip above it, with no surge and no meter, feeds the Bay A and desk gear, the lid monitors, the laptops and the POWER > WING outlet. The UPS side carries 13 loads, an estimated 275 to 465W.

**Why it's there.** One UPS backs both the booth and the Wing. The Wing case can't carry its own UPS, so this is the only way to back the Wing.

**Why not another.** The ProX X-PC10ASTRO strip was passed on (section 3).

**Cost.** About $435, mostly estimated.

**Status.** Decided. The CPS1215RM listing still needs confirming (section 4).

### Recorder and Dante satellite (U10)

**What it does.** The HyperDeck Studio HD Mini records the clean feed from ATEM AUX. A new 8-port PoE switch is the Bay A Dante satellite. It takes the DANTE TRUNK, powers the AVIO and connects the ProPresenter Mac's Dante adapter. Both share one 1U vented shelf.

**Why it's there.** Every network jumper stays in Bay A, and one shelf model is the cheapest mount.

**Why not another.** An Etsy dual mount and a Teranex shelf were both dropped (section 3).

**Cost.** The HD Mini is owned. The PoE switch carries a placeholder of about $20 until its model is priced (section 4). The shelf is in the small-parts card.

**Status.** Decided. The PoE model is open, and shelf fit gets checked on delivery.

### Owned gear moved into Bay A

**What it does.** Resi takes SDI from ATEM OUT 8 and streams out through the router. The LED wall transcoder takes HDMI from ProPresenter and sends two Ethernet runs to LED 1 and LED 2. The TL-SG105 production switch and the AX21 router run the production network, and wired internet from the venue or Starlink connects to the router. Dante stays on its own isolated network.

**Why it's there.** It's the gear we run today, moved into the rack, and every network jumper stays in Bay A.

**Cost.** Already owned.

**What we give up.** Every wired production port is used, so adding a device later means a bigger switch.

**Status.** Decided. The Bay A 4U top is left open.

### Small parts: shelves, cable management, screws and jumpers

**What it does.** This covers the vented shelves, a lacing bar, hook-and-loop straps, rack screws and cage nuts, short power cords, color-coded patch cables and short HDMI jumpers.

**Why it's there.** The lacing bar holds the cable runs, power on one side and signal on the other, and velcro straps keep the rest tidy.

**Why not another.** Zip ties were ruled out in favor of hook-and-loop.

**Cost.** About $320, estimates.

**Status.** The parts list is decided. The screw thread and built-in rails are open (section 4).

### Cooling: 1U thermostatic fan and vented blanks

**What it does.** A 1U thermostatic fan at the top of Bay A exhausts toward the open back. Perforated blanks fill the reserve spaces.

**Why it's there.** The open back does most of the cooling, and the heat load is low. The fan is insurance for hot outdoor events, and the blanks stop intake air from short-circuiting around the gear.

**Why not another.** Open gaps were rejected in favor of blanks (section 3).

**Cost.** About $190, estimates.

**What we give up.** At a venue the open back has to stay clear, so the case can't be pushed against a wall.

**Status.** Decided.

## d. Bay B, storage and charging

Bay B charges the camera batteries. It carries no signal. The rest of it is storage, and all cabling is stored with the two cases.

### Bay B power: Clear Power CPPDU511 + Tripp Lite RS1215-RA

**What it does.** The metered CPPDU511 at the front lights the drawers. The RS1215-RA at the rear feeds the camera chargers and plugs into the CPPDU511, so both read on one meter. Power drop 2 feeds them.

**Why it's there.** The front lights the drawers, the rear outlets face the chargers, and there's one meter.

**Cost.** About $245, one firm price and one quoted.

**What we give up.** This side isn't battery-backed, so nothing that must stay on plugs in here.

**Status.** Decided. Dropping the RS1215-RA is an open proposal (section 4).

### Pass-through to the separate charging station

**What it does.** The CPPDU511 passes power through an extension to a quad box at a separate charging station. One of the removable lids, popped up as a table, could hold that charging area.

**Why it's there.** It gives a separate charging area its own outlets.

**Cost.** About $90 for the quad box. The extension isn't costed.

**Status.** The quad box is decided. The extension details are open.

### Storage: two 4U drawers

**What it does.** Two 4U drawers fill the lower half of Bay B, with heavier items in the lower drawer to balance the case. All cabling is stored with the two cases, not elsewhere. The upper spaces and Top B are left open, with room for a bigger switch or the Mac Minis later.

**Why it's there.** Bay B is the storage side of the rack.

**Cost.** About $310, an estimate.

**Status.** Decided.

## e. The rear panel

This is the only face anyone touches during setup. Anyone setting up matches the color and number on each cable to a jack, and plugs in. Nothing plugs into a device directly, so field wear never reaches the gear.

**Panel 1: video**

| CAM 1 BNC | CAM 2 BNC | CAM 3 BNC | CAM 4 BNC | CAM 5 BNC | CAM 6 BNC | CAM 7 BNC | CAM 8 BNC | CONF OUT HDMI | blank - | blank - | blank - |
|---|---|---|---|---|---|---|---|---|---|---|---|

**Panel 2: data, audio and power**

| DANTE TRUNK etherCON | LED 1 etherCON | LED 2 etherCON | INTERNET IN etherCON | PROD TRUNK etherCON | TALKBACK XLR-F | IEM XLR-M | SPARE etherCON | blank - | blank - | blank - | POWER > WING powerCON |
|---|---|---|---|---|---|---|---|---|---|---|---|

Panel 3 is a spare row of 12 holes. The panel layout is decided. All eight camera jacks stay on the panel, so every switcher input is visible. Today ATEM IN 2 is the Hollyland RX and IN 8 is the UltraStudio. With the Hollyland on the flip-up lid, its SDI still has to reach IN 2, and one option is to run it through the CAM 2 jack. How both connect is open and may change.

### One panel, three plates

**What it does.** Three Neutrik D-series 12-hole plates sit on the Bay A rear rails. Panel 1 is video, panel 2 is data, audio and power, and panel 3 is a spare row.

**Why it's there.** It gives us one place to connect, and field wear never reaches the gear.

**Cost.** About $60, an estimate.

**Status.** Decided, including the panel order and labels.

### The connectors

**What it does.** There are eight BNC camera jacks, an HDMI confidence output, six etherCON jacks, a female TALKBACK XLR, a male IEM XLR and a powerCON outlet for the Wing. Blanking plates fill the empty holes. The panel note above covers how the camera jacks map to the switcher.

**Why it's there.** A mic plugs into the talkback jack, the IEM jack feeds a pack, and power flows from the AVPRO to the Wing.

**Why not another.** A CAT6A etherCON cable connector was dropped because it doesn't mate with the panel jacks.

**Cost.** About $240, estimates.

**Status.** The layout is decided. The powerCON and etherCON parts are open until they're checked (section 4).

### Cable ends, jumpers and spares

**What it does.** etherCON cable carriers and RJ45 plugs fit the trunk, LED and internet ends. powerCON cable ends make the Wing power link. SDI jumpers run from the panel to the switcher, and loose spares cover both panels.

**Why it's there.** Each carrier end gets a fresh crimp, and spares cover both panels.

**Cost.** About $250, estimates.

**Status.** The carrier count is bought high. The internet drop's far end is open (section 4).

### Color code and labels

**What it does.** Every cable has a color and a number, and the jack it lands on has the same color and number. Blue is camera SDI, green is Dante, yellow is production network, red is internet, purple is LED data, white is HDMI, orange is XLR and AES50, and black is power.

**Why it's there.** Anyone setting up matches colors, not port names. Green against yellow is the split that matters most.

**Cost.** Covered by the straps and labeling in the small-parts card.

**Status.** Decided.

## f. The five short cables between the cases

At a service or event the two cases sit side by side, and five short cables join them. That's the whole connection between the AVPRO and the Wing case.

### Two data trunks and two audio lines

**What it does.** The PROD trunk carries Wing control from the router. The DANTE trunk joins the two Dante switches. Both are 15ft shielded Cat6A with etherCON ends. TALKBACK carries Bianca's mic to Wing In 8, and IEM carries Wing Out 8 back to her pack, both on 15ft XLR.

**Why it's there.** The cases always sit side by side, so short cables are enough. The AES50 to the stage box runs stage-side.

**Cost.** About $25 for the trunk cables, an estimate. The XLR cables are owned.

**What we give up.** If a venue ever needs the cases apart, we'd add longer trunks then.

**Status.** Decided. Wing In 8 and Out 8 are closed.

### The Wing power link

**What it does.** A 15ft powerCON TRUE1 cable runs from the AVPRO's POWER > WING outlet to the Wing.

**Why it's there.** It carries battery-backed power from the AVPRO's UPS to the Wing.

**Why not another.** The 12/3 SOOW used for the wall drops is too thick for the TRUE1 cable ends (section 3).

**Cost.** About $15, an estimate.

**Status.** Decided. The matched TRUE1 family is open until it's confirmed.

## g. The Wing case

At a service or event the Wing is operated from its case and records to the ProTools Mac over USB-C. The ProTools Mac handles both the live event mix and the stream mix, so the separate stream-mix laptop is no longer needed.

**Wing panel (if the case can take one)**

| CONTROL<br>etherCON | DANTE TRUNK<br>etherCON | AES50<br>etherCON | TALKBACK<br>XLR-M | IEM<br>XLR-F | POWER IN<br>powerCON | blank<br>- | blank<br>- |
|---|---|---|---|---|---|---|---|

Otherwise cables route out the side of the case.



The BXWING product listing photo.

### The case: ProX BXWING hydraulic flip case

**What it does.** It holds the Behringer Wing beside the AVPRO.

**Why it's there.** Today the Wing travels in a wheeled case that has to be set on a table before we can use it. The BXWING's hydraulic lid flips the Wing up into its working position, so it doesn't need a table at all. Removing that table is one of the main reasons for this upgrade.

**Cost.** About $900, a placeholder until Matt's quote.

**What we give up.** None recorded.

**Status.** The design is decided. The price is open until Matt's quote.

### Inside: the Wing, its Dante switch and the ProTools Mac

**What it does.** The Wing is owned. The owned 5-port TL-SG1005P is the Wing case Dante switch, connecting the Wing's Dante card, the trunk and the ProTools Mac, three of five ports. The ProTools Mac also has USB-C to the Wing for recording.

**Why it's there.** Three ports fit the 5-port switch, and the new PoE switch goes to Bay A to power the AVIO.

**Why not another.** An 8-port switch here was planned first (section 3).

**Cost.** Already owned.

**Status.** Decided. The ProTools Mac recording path is open (section 4).

### Backed by the AVPRO's UPS

**What it does.** The Wing runs on battery-backed power from the AVPRO over the Wing power link.

**Why it's there.** The Wing case can't carry a UPS, so this is the only way to back the Wing.

**Cost.** Included in the UPS line.

**What we give up.** For standalone use the Wing goes on a wall outlet.

**Status.** Decided.

### The Wing panel

**What it does.** If the case can take it, a small D-series plate gives the five cables and the AES50 their own jacks. The genders mirror the AVPRO panel, and a power inlet feeds the Wing.

**Why it's there.** It gives the Wing case the same plug-in panel as the AVPRO.

**Why not another.** If the case can't take a panel, the connections run out the side of the case.

**Cost.** About $100, not in the first order.

**Status.** Conditional, and likely doable. I'll inspect the case on delivery.

## h. Power from the wall

At a service or event two long cords run from wall outlets to the rig. The first feeds the AVPRO's battery backup, and the Wing through it. The second feeds Bay B and the charging station.

### Two 125ft power drops

**What it does.** Each drop is 125ft of 12/3 SOOW with a plug at the wall and a connector body at the rig. Drop 1 feeds only the UPS. Drop 2 feeds the Bay B CPPDU511, which feeds the charging station. The two are built the same, so either can stand in for the other.

**Why it's there.** Venue distances vary widely, and long is the safe default.

**Cost.** About $470, estimates.

**Status.** Decided. The plug and connector parts are assumed, and an equal 15A industrial part can be substituted.

### Circuits

**What it does.** The whole rig draws well under one 15A circuit, so one works. A second circuit is preferred, not required. When a venue has two, each drop goes on a different circuit. We never tie circuits together.

**Why it's there.** We can't know a venue's circuit count ahead of time.

**Cost.** No cost.

**Status.** Decided.

### i. Field cables

At a service or event the field cables are the runs out into the room: cameras, the LED wall, the confidence monitor and internet. Each one is labeled and color-coded, and each lands on the rear panel. All field cabling is stored with the two cases.

#### Camera SDI

**What it does.** Each wired camera sends HDMI to its BMD converter, which turns it into 1080p SDI. The SDI runs back over coax to the panel and on to the ATEM at the case. We already own all the long SDI runs and spools, and they're 100ft. We run three cameras at every event, two Pocket 4K bodies and the DJI, and more only when we rent them.

**Why it's there.** The cameras are HDMI out and the switcher is SDI in. Locking, rugged coax handles long runs in a spread-out room.

**Why not another.** An HDMI-to-converter box near the cameras wasn't chosen, because HDMI doesn't carry far enough.

**Cost.** Already owned, converters and cables.

**What we give up.** Each converter runs on its own adapter, so every wired camera position needs mains power.

**Status.** Decided: we implement with the SDI we own. Whether to switch to AOC is open (section 4).

#### The camera-cable trial

**What it does.** Emily runs the SDI cables we already own over a real deploy. The trial checks pull-through, cable and connector reliability, and picture over length. It doesn't test camera control.

**Why it's there.** It checks the owned cables before we rely on them.

**Cost.** No cost.

**Status.** Open. No date is set.

#### AOC, an open decision

**What it does.** AOC would replace the SDI coax on the camera runs. The camera's HDMI would go straight into the cable, so there's no BMD converter at each camera, and the converters could live in the Wing case.

**Why it's there.** AOC is lighter and more flexible than coax, and it runs longer distances without the signal degrading.

**Cost.** About $150 for a 50m run on a spool, an estimate from my notes with Matt. It isn't in the BOM.

**Status.** Open decision. We implement with the owned SDI until it's made.

#### LED wall, confidence monitor and internet runs

**What it does.** The LED wall takes two Ethernet runs from LED 1 and LED 2. The confidence monitor takes HDMI from CONF OUT. Internet comes from the venue or Starlink into INTERNET IN.

**Why it's there.** Each field run lands on its own labeled jack.

**Cost.** Today's LED and internet runs are already owned. They get etherCON carriers at the panel end.

**Status.** Decided. The LED wall receiver takes standard Ethernet.

## 3. What we chose not to do

These are the options build/ records as passed on, dropped or parked, with the reason given at the time.

| | |
|---|---|
| **The ProX X-PC10ASTRO power strip** | Stacked surge behind a UPS isn't good practice, and the metered CPPDU511 in Bay B already gives us a meter. |
| **An Etsy dual switch mount and the Teranex Mini Rack Shelf** | One 1U vented shelf model does the job, keeps every network jumper in Bay A and costs the least. |
| **Rack mounts for the converters** | The converters live at the cameras. |
| **An HDMI-to-converter box near the cameras** | It's on record but not chosen. HDMI distance is the catch. |
| **A 2000-nit field-monitor tier** | It was replaced by the LG on cost. Brightness is open again, so a brighter or used option may come back (section 4). |
| **Three networks** | We have two physical networks. The third was a zoning aid. |

| | |
|---|---|
| **A CAT6A etherCON cable connector** | It belongs to a different family and doesn't mate with the panel jacks. |
| **An 8-port switch in the Wing case** | The Wing case uses three ports, so the owned 5-port TL-SG1005P is the right size there. The 8-port PoE switch goes to Bay A instead. |
| **12/3 SOOW for the Wing power link** | It's too thick for the TRUE1 cable ends. |
| **A surge strip behind the UPS** | No surge behind the UPS. The laptops plug straight into the battery-backed CPS1215RM instead. |
| **Zip ties** | Hook-and-loop and the lacing bar are used instead. |
| **Open gaps in reserve rack spaces** | Open slots let intake air short-circuit around the gear, so they get vented blanks. |
| **Camera control in this build** | It's parked. The owned converters are bidirectional, so it needs only a return SDI per camera later. |
| **Stage power in this BOM** | It's a separate subsystem and doesn't gate the order. |
| **Using the AVPRO side compartments** | The design doesn't use them. That gets confirmed on delivery. |

# 4. What we still don't know

Every item here is open unless it says otherwise. Owners and dates are in the appendix.

### What affects the order

- The case and freight price. The $1,650.00 in the total is a placeholder until Matt's quote, and the order can't go in without it. The spend ceiling for that quote is also open. Bianca and I set it before we walk through this with Sunny.
- The connector families. The powerCON and etherCON parts get checked against Neutrik's tables before those lines are ordered.
- The CPS1215RM listing. I need to confirm the Amazon listing is the right strip. It affects that line only.
- The rack rails and screws. The screws are M6 or 10-32, and that needs confirming. It's also open whether the case comes with rails built in or they're added after purchase. If Matt doesn't know, this waits until the case arrives.

### What affects the cost

- Field camera cable isn't a cost. We already own all the long SDI runs and spools, at 100ft. Any power, HDMI or data field cables we add later should be 125ft or more.
- Whether to switch field camera cable to AOC is an open decision. AOC runs about $150 for a 50m run on a spool, and it isn't in the BOM.
- The new 8-port PoE switch for Bay A. The BOM carries $19.99 as a placeholder, and a PoE model will cost more than that.
- The extension from Bay B to the charging station isn't costed. Its details are open.
- The count of etherCON cable ends depends on the internet drop's far end. The BOM already buys the higher count.

- Possible: drop the RS1215-RA ($94.98) and feed the charging table by a powerCON jumper from the CPPDU511. The jumper and panel outlet aren't specced yet.
- The Hollyland RX lid mount isn't specced or costed yet.
- The ATEM retention tray isn't costed. We need to build a foam or plywood tray, or secure the ATEM another way, and Matt may have suggestions.
- Lid monitor brightness. The 300-nit LG is the minimum worth considering and probably isn't bright enough under a tent. I need to research brighter screens or find a used option, and the recorded fallback is a 1000-nit unit for the LEFT lid only. Brighter screens cost more, so this line may go up.
- Lid legs. I need Matt to confirm whether legs come on the two removable AVPRO lids. If they don't, we add them as a line item. It doesn't gate the order.
- Matt's "regret ask" covers long-lead items, freight consolidation and anything hard to add later. It comes with his quote.

### What affects the Wing case

- Whether the Wing case can take a connector plate. I'll inspect it on delivery. If it can, we buy the $100.00 Wing block. If it can't, the connections run out the side of the case.
- The ProTools Mac recording path has to be settled for the case layout. The Mac needs room for both its USB-C and Dante cables.

A few smaller checks don't touch any of these. They cover shelf, monitor and side-door fit on delivery, two equipment labels, the drum-kit Dante question, where the UltraStudio mounts (rack-mounted, near the Mac Mini, or the desk near the ATEM), and how the Hollyland RX connects to ATEM IN 2.

## 5. What it costs

The first order is about $4,600 ($4,641.84). Each category below has one line on what it covers.

| Category | Cost | What it covers and where it stands |
|---|---|---|
| Cases and freight | $1,650.00 | OPEN Matt's estimates for both cases and consolidated freight. The real number comes from his quote. |
| Panel and connectors | $473.41 | ESTIMATE The rear panel plates and jacks, loose spares and the cable ends that fit them. |
| Cabling | $276.10 | ESTIMATE Short internal jumpers, the two data trunks and the short Wing power link. |
| Power | $1,241.34 | ESTIMATE QUAD BOX: DECIDED The UPS and its strip, the Bay B strips, the two wall drops and the charging quad box. |
| Cooling and mounting | $661.00 | ESTIMATE The fan and blanks, three shelves, two drawers, cable management and screws. |
| Network | $19.99 | PLACEHOLDER The new 8-port PoE switch for Bay A. The figure is a placeholder, because the PoE model isn't priced. |
| Monitors | $320.00 | ESTIMATE Two lid monitors and their right-angle adapters. |
| **First order total** | **$4,641.84** | |
| Conditional items | $100.00 | CONDITIONAL The Wing panel parts, bought only if the Wing case can take a panel. If it can't, the connections run out the side of the case, and a TRUE1-to-IEC jumper (not costed) carries the Wing power link to the Wing's IEC inlet. |
| **All lines** | **$4,741.84** | |
| Already owned | not in any total | Almost all the gear inside. |
| Lid legs | not costed | OPEN Open until Matt confirms whether they come with the lids. |

## 6. How it rolls out and the risks

The owned gear moves into the cases in stages, starting from a setup closest to today's. Each stage is proved before the next. That's decided.

Weather is an accepted risk, because we often set up outside with no cover. The plan is a fitted rain cover or poly sheet and clamps in the case, with a two-minute cover-and-kill routine. That plan is still open, and I'll have it in place before the first live deploy.

# Appendix

## Full BOM (BOM v7, 2026-09-28)

Owned gear isn't in any total. Not in this BOM: camera field SDI cable (already owned, 100ft runs), stage power, and the XLR trunk cables (owned). "est" means an estimated price. * BOM v7 lists the quad box as "tentative, may drop (N19)". It's shown as decided.

| Group | Item | Model / SKU | Qty | Unit | Ext | Price status |
|---|---|---|---|---|---|---|
| Cases | Rolling dual-bay workstation | ProX XZF-AVPRO2X12U2X4U | 1 | $500.00 | **$500.00** | placeholder, to verify (Matt) |
| Cases | Wing hydraulic flip case | ProX XZF-BXWING | 1 | $900.00 | **$900.00** | placeholder, to verify (Matt) |
| Cases | Freight, both cases (consolidated) | - | 1 | $250.00 | **$250.00** | placeholder, to verify (Matt) |
| AVPRO panel | 1U 12-hole D-series plate | My Cable Mart D-series | 3 | $20.00 | **$60.00** | est |
| AVPRO panel | BNC 75 ohm feedthrough (camera SDI) | Neutrik NBB75DFI | 8 | $8.00 | **$64.00** | est |
| AVPRO panel | HDMI feedthrough | Neutrik NAHDMI-W-B | 1 | $25.00 | **$25.00** | est |
| AVPRO panel | etherCON feedthrough | Neutrik NE8FDP-B | 6 | $12.00 | **$72.00** | est |
| AVPRO panel | XLR female chassis | Neutrik NC3FD-L-B-1 | 1 | $6.00 | **$6.00** | est |
| AVPRO panel | XLR male chassis | Neutrik NC3MD-L-B-1 | 1 | $6.00 | **$6.00** | est |
| AVPRO panel | powerCON TRUE1 chassis outlet | Neutrik NAC3FPX (family TBC, D7) | 1 | $20.00 | **$20.00** | est |
| AVPRO panel | Feed cord for the POWER > WING outlet | 12/3 cord with 5-15P, wired to the outlet | 1 | $12.00 | **$12.00** | est |
| AVPRO panel | D-hole blanking plate | Neutrik DBA-BL | 18 | $2.00 | **$36.00** | est |
| Spares | BNC feedthrough, spare | Neutrik NBB75DFI | 2 | $8.00 | **$16.00** | est |
| Spares | etherCON feedthrough, spare | Neutrik NE8FDP-B | 2 | $12.00 | **$24.00** | est |
| Spares | XLR female chassis, spare | Neutrik NC3FD-L-B-1 | 1 | $6.00 | **$6.00** | est |
| Spares | XLR male chassis, spare | Neutrik NC3MD-L-B-1 | 1 | $6.00 | **$6.00** | est |
| Cable ends | etherCON cable carrier (standard etherCON, over an RJ45 plug) | Neutrik NE8MX-1 (TBC, D7) | 11 | $3.31 | **$36.41** | est |
| Cable ends | RJ45 plugs, shielded (pack), for the carrier ends | Shielded Cat6A RJ45 plugs, 10-pk | 1 | $12.00 | **$12.00** | est |
| Cable ends | powerCON TRUE1 cable connector, male | Neutrik NAC3MX-W (TBC, D7) | 2 | $18.00 | **$36.00** | est |
| Cable ends | powerCON TRUE1 cable connector, female | Neutrik NAC3FX-W (TBC, D7) | 2 | $18.00 | **$36.00** | est |
| Power | UPS, 2U pure sine (backs booth + Wing) | CyberPower CP1500PFCRM2U | 1 | $375.00 | **$375.00** | est |
| Power | Rack PDU from the UPS (basic, no surge) | CyberPower CPS1215RM | 1 | $59.99 | **$59.99** | firm; listing check (G5) |
| Power | Bay B front PDU, metered, drawer lights | Clear Power CPPDU511 | 1 | $149.99 | **$149.99** | firm |
| Power | Bay B rear strip, camera chargers | Tripp Lite RS1215-RA | 1 | $94.98 | **$94.98** | quoted 2026-08-20 |
| Cooling | 1U thermostatic exhaust fan | AC Infinity CLOUDPLATE T1 | 1 | $150.00 | **$150.00** | est |
| Cooling | 1U perforated blank panel | Steel perforated (Amazon) | 3 | $12.00 | **$36.00** | est |
| Mounting | 1U vented cantilever shelf, 10in deep | AC Infinity 1U vented (B01C9KYUG8) | 3 | $25.00 | **$75.00** | est |
| Network | Bay A Dante satellite switch, 8-port PoE | PoE model open | 1 | $19.99 | **$19.99** | placeholder: PoE model not priced |
| Storage | 4U rack drawer | StarTech 4UDRAWER | 2 | $155.00 | **$310.00** | est |
| Cable mgmt | 1U cable lacing bar | StarTech 12S-CABLE-LACING-BAR | 1 | $25.00 | **$25.00** | est |
| Cable mgmt | Hook-and-loop straps | ONE-WRAP + WrapStrap | 1 | $40.00 | **$40.00** | est |

| Group | Item | Model / SKU | Qty | Unit | Ext | Price status |
|---|---|---|---|---|---|---|
| Hardware | Rack screws + cage nuts | M6 or 10-32 (open; check the rails) | 1 | $25.00 | **$25.00** | est; waits on G6 |
| Cabling | SDI BNC jumpers, panel to ATEM | Belden 1855A / premade BNC-BNC | 8 | $10.00 | **$80.00** | est |
| Cabling | Cat6A patch, color-coded (lot) | Monoprice SlimRun Cat6A 10-pk | 1 | $80.00 | **$80.00** | est |
| Cabling | Trunk cable, shielded Cat6A, 15ft | Shielded Cat6A patch, 15ft | 2 | $12.00 | **$24.00** | est |
| Cabling | HDMI 2.0 jumper, short | HDMI 2.0, 6-10ft | 3 | $12.00 | **$36.00** | est |
| Cabling | Short power cord, 5-15P to IEC C13 | 2-3ft | 6 | $7.00 | **$42.00** | est |
| Cabling | Umbilical power cable (AVPRO > Wing), 14/3 SJOOW | 14/3 SJOOW, cut to length (ft) | 15 | $0.94 | **$14.10** | est |
| Venue power | Power drop cable, 12/3 SOOW (2 x 125ft) | 12/3 SOOW, cut to length (ft) | 250 | $1.63 | **$407.50** | est |
| Venue power | Edison plug, 5-15P (wall end) | Hubbell HBL5266C | 2 | $14.00 | **$28.00** | est |
| Venue power | Edison connector body, 5-15R (rig end) | Hubbell HBL5269C | 2 | $18.00 | **$36.00** | est |
| Venue power | Quad box, 4 outlets, 2ft 12/3 tail with 5-15P | BarnDoor QB-M-2FT metal quad box | 1 | $89.88 | **$89.88** | decided * |
| Monitors | Lid monitor, 24in matte IPS | LG UltraGear 24GS60F-B | 2 | $150.00 | **$300.00** | est |
| Monitors | Right-angle HDMI + DC adapters (lot) | 90-deg HDMI + DC barrel | 1 | $20.00 | **$20.00** | est |
| **First order total** | | | | | **$4,641.84** | of which $1,650.00 case + freight placeholders |
| **Wing block (conditional, only if the BXWING can take a panel, N1)** | | | | | | |
| Wing panel | Small D-series plate (8 holes) | D-series plate | 1 | $20.00 | **$20.00** | est |
| Wing panel | etherCON feedthrough | Neutrik NE8FDP-B | 3 | $12.00 | **$36.00** | est |
| Wing panel | XLR male chassis | Neutrik NC3MD-L-B-1 | 1 | $6.00 | **$6.00** | est |
| Wing panel | XLR female chassis | Neutrik NC3FD-L-B-1 | 1 | $6.00 | **$6.00** | est |
| Wing panel | powerCON TRUE1 chassis inlet | Neutrik NAC3MPX (family TBC, D7) | 1 | $20.00 | **$20.00** | est |
| Wing panel | Inlet-to-Wing power pigtail | IEC C13 cord, cut and wired to the inlet | 1 | $8.00 | **$8.00** | est |
| Wing panel | D-hole blanking plate | Neutrik DBA-BL | 2 | $2.00 | **$4.00** | est |
| Wing panel | FALLBACK ONLY: powerCON TRUE1 to IEC C13 jumper | TRUE1 male to C13, short | 0 | $30.00 | **$0.00** | est; qty 0. Only if the connections run out the side of the case: the Wing power link then needs it to reach the Wing's IEC inlet |
| **Wing block total** | | | | | **$100.00** | |
| **All lines total** | | | | | **$4,741.84** | |

## Locked decisions (DESIGN section 5)

All are decided. Each one changes only through a written proposal that I accept.

| # | Decision | Reason |
|---|---|---|
| **1** | We implement with the SDI we already own: a BMD converter at each camera, 1080p, coax to the panel. Whether to switch to AOC is open. | HDMI out, SDI in; locking, rugged and already owned |
| **2** | Owned SDI field runs are 100ft. Any power, HDMI or data field cables we add should be 125ft or more. | Venue distances vary widely; long is the safe default |
| **3** | Three cameras at every event: two Pocket 4K + DJI; more only when rented | That's what runs today (D9) |
| **4** | One rear I/O panel, three Neutrik D-series 12-hole plates on Bay A | One touchpoint; field wear never reaches the gear |
| **5** | Panel order and labels (POWER > WING last, TRUNK names) | The fabrication drawing (D13) |
| **6** | Genders: AVPRO TALKBACK female, IEM male, POWER an outlet; the Wing panel mirrors them | A mic plugs into the talkback jack; the IEM jack feeds a pack; power flows AVPRO → Wing |
| **7** | ATEM IN 2 is always the Hollyland RX, and IN 8 is always the UltraStudio. How each connects is open. | Matches the running ATEM |
| **8** | Color code (BUILD-SPEC §12); green vs yellow is the split that matters | Anyone setting up matches colors, not port names |
| **9** | Labeling: "match the color, match the number" | Anyone setting up matches colors, not port names |
| **10** | One UPS (CP1500PFCRM2U) backs the booth and the Wing over the umbilical | The BXWING can't carry a UPS; this is the only way to back the Wing |
| **11** | No surge behind the UPS; the ProX X-PC10ASTRO was passed on | Stacked surge behind a UPS isn't good practice, and Bay B's metered CPPDU511 already gives us a meter |
| **12** | Runs on one 15A circuit, two preferred, never tied. Power arrives on two 125ft 12/3 SOOW drops with Edison ends: drop 1 → AVPRO (and the Wing through it), drop 2 → the charging station | The whole rig draws well under 15A; venue circuit counts are unknowable (D20) |
| **13** | Bay B power: CPPDU511 front (metered, drawer lights) + RS1215-RA rear (camera chargers), chained, utility side | Front lights the drawers; rear outlets face the chargers; one meter (D3) |
| **14** | The desk laptops plug into the battery-backed CPS1215RM strip. Decided. | They keep running if venue power drops |
| **15** | Switches: owned TL-SG1005P (5-port) = Wing case Dante switch, 3 ports used; owned TL-SG105 = Bay A production switch; new 8-port PoE switch = Bay A Dante satellite at U10, powers the AVIO | The AVIO needs PoE in Bay A, and the Wing case needs only 3 ports |
| **16** | Bay A mounting: Resi U11; HD Mini + new 8-port PoE switch U10; TL-SG105 + AX21 U8; all on the same 1U vented shelf; Etsy dual mount and Teranex shelf dropped | Every network jumper stays in Bay A; cheapest shelf (D5, D19) |
| **17** | Two networks, not three: Dante (isolated) and Production; internet is the AX21's WAN | That's the physical wiring; "three" was a zoning aid |
| **18** | Router is the AX21 (older docs say AX55 for the same router) | D11 |
| **19** | UltraStudio placement is open: rack-mounted or near the Mac Mini, or the desk near the ATEM. The Hollyland RX mounts on top of the flip-up lid when open and stows in a drawer or with the cameras until setup. The Bay A 4U top is left open. | The receiver won't get good signal inside the rack |

| # | Decision | Reason |
|---|---|---|
| **20** | Lid monitors: 2 × LG 24GS60F-B, 300 nit matte. It may not be bright enough even under a tent or canopy, and direct sun needs angling. Model reopened for brightness research | D14; right-angle adapters for lid clearance |
| **21** | LEFT lid = ATEM multiview day one; RIGHT lid = the ProPresenter Mac's second screen, ready for a later move to a Mac Mini | Multiview can't depend on a laptop |
| **22** | Cooling: open back primary; 1U thermostatic fan at U12 exhausting to the back; vented blanks in reserve U's | Low heat load; blanks stop airflow short-circuits |
| **23** | Umbilical = 5 short cables: PROD, DANTE, TALKBACK, IEM, POWER (15ft powerCON TRUE1 link, 14/3 SJOOW) | The cases sit side by side; TRUE1 cable ends take 6-12mm cable (D20). AES50 is stage-side |
| **24** | All five trunks are short; the cases always sit side by side | Q3, D20. Longer runs can be added later |
| **25** | The Wing panel is conditional: planned, pending BXWING inspection. If the case can't take one, the connections run out the side of the case | Nobody has confirmed the case takes a panel, but it's likely doable |
| **26** | Stage power is out of the build BOM | Separate subsystem; doesn't gate the order (D8) |
| **27** | Case + freight figure comes from the BOM ($1,650), labeled placeholder until Matt quotes | One number everywhere (D6) |

## All open items

Running-config items are left out. G3 and G4 closed on 2026-09-28. Stage power items (N17, N18) are tracked with the separate stage-power project.

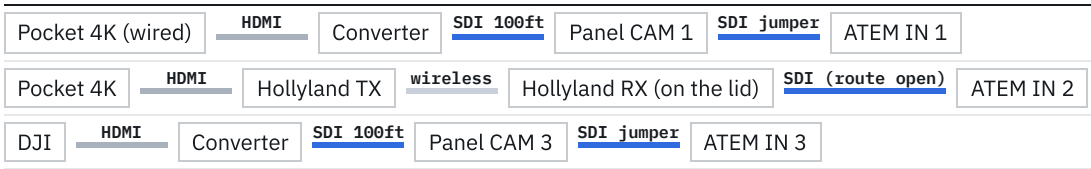
| Item | Owner | Due | Gates the order | Status |
|---|---|---|---|---|
| **G1 Case and freight quote:** AVPRO price, BXWING price, consolidated freight for both (closed AVPRO is 51.5 × 48.2 × 31.6in). BOM placeholders: $500 / $900 / $250 | Matt | 2026-10-19 | Yes: the order gate | OPEN |
| **G2 One matched powerCON TRUE1 family:** AVPRO chassis outlet, both umbilical cable ends, and the Wing chassis inlet (only if the Wing panel happens). Plus the etherCON pair: NE8FDP-B panel jacks with NE8MX-1 carriers (standard etherCON), checked against the Neutrik compatibility table before the BOM calls them firm | Matt / vendor; Sam checks the datasheets | 2026-10-19 | Yes, for the connector lines | OPEN |
| **G5 CPS1215RM listing:** confirm Amazon B00077IG3O is the CPS1215RM | Sam | 2026-10-19 | That line only | OPEN |
| **G6 Rack rails and thread:** tapped 10-32, or square-hole for M6 cage nuts? Both are open. Also confirm whether the AVPRO comes with rails built in or they're added after purchase. Matt may know; otherwise it's checked at delivery | Sam (ask Matt first) | on case delivery (assumed by 2026-11-15) | Screw line only. Order it after delivery if Matt doesn't know | OPEN |
| **N1 BXWING inspection: can it take a panel-mount D-series plate?** Yes → buy the Wing block. No → the connections run out the side of the case. | Sam (ask Matt first) | on case delivery (assumed by 2026-11-15) | Wing block only | OPEN WING BLOCK |
| **N8 Camera-cable trial:** Emily runs the SDI cables we already own (100ft) over a real deploy. Checks pull-through, cable and connector reliability, and picture over length. No camera control.<br>> *Whether to switch field camera cable to AOC is an open decision. AOC would let the BMD converters live in the Wing case.* | Emily (trial); Sam, with Matt (spec and cost) | No date set | No | OPEN |
| **N15 Measure today's setup and teardown time** before the cutover, and again after, to test the ~40 → ~15 connections and ~30 minutes each way estimates | Sam | 2026-10-04, then at the first event after cutover | No | OPEN |

| Item | Owner | Due | Gates the order | Status |
|---|---|---|---|---|
| **N16 Weather: accepted risk.** Setups are often outside with no cover. Keep a fitted rain cover or poly sheet + clamps in the case and a two-minute cover-and-kill plan. A powered open-back rack in sudden rain is the one fast path to a big loss | Sam | before the first live deploy (assumed by 2026-11-15) | No | RISK ACCEPTED PLAN: OPEN |
| **N19 Where is the AV charging station?** Drop 2 feeds the Bay B CPPDU511, which passes power through to a separate charging station on a quad box. The quad box is in the first order. Details of the extension are open. A second wall circuit is preferred, not required. | Sam | Closed | No | DECIDED |
| **N6 Wing talkback and IEM:** Wing XLR In 8 is Bianca's talkback; Out 8 is her IEM, per the ELIM Production App. | Tyler | Closed | No | CLOSED |
| **N13 Streaming laptop** The app's "video stream laptop" and the switcher laptop are the same machine, now called the streaming laptop. | Sam | Closed | No | CLOSED |
| **G7 Laptop power** The desk laptops plug into the battery-backed CPS1215RM strip. The separate laptop strip is dropped from the BOM. | Sam | Closed | No | CLOSED |
| **N2 M7: LED wall and internet ends** The LED wall receiver takes standard Ethernet. The internet drop's far end is still open. It sets the etherCON end count; the BOM already buys the high count. | Sam | 2026-10-04 (next service) | No | OPEN |
| **N3 M3: side compartments** Do the AVPRO side doors open to usable storage, and what are the inside dimensions? The design doesn't use them. | Sam (ask Matt/ProX) | On case delivery (assumed by 2026-11-15) | No | OPEN |
| **N4 U10 shelf fit** The 10in AC Infinity shelf takes the HD Mini and the new 8-port PoE switch side by side. | Sam | On shelf delivery (assumed by 2026-10-15) | No | OPEN |
| **N5 M5: monitor fit** LG 24GS60F-B panel height without stand + right-angle adapter depth vs the 16.1in lid; two widths vs 49.6in. | Sam | On case delivery (assumed by 2026-11-15) | No | OPEN |
| **N7 ProTools Mac recording path** Wing USB-C to the ProTools Mac, inside the BXWING, settled for the case layout. The ProTools Mac is also a Dante endpoint on the Wing case switch, so the case needs room for both cables. | Tyler | 2026-10-04 | No | OPEN |
| **N10 M2: AVIO label** Read the AVIO model label (AVIOAO2 per the port map). It's confirmed PoE-powered. | Sam | 2026-10-04 | No | OPEN |
| **N11 R-019: drum kit Dante** Does Omar have a separate Dante device at the drum kit? (yes/no) | Sam | 2026-10-04 | No | OPEN |
| **N12 R-024: Shure combiner label** Read the Shure transmit combiner label (UA844+SWB? typed "ua344"). | Sam | 2026-10-04 | No | OPEN |
| **N14 The "regret ask"** Long-lead items, freight consolidation, anything hard to add later. | Matt | 2026-10-19, with the quote | No | OPEN |
| **Lid legs:** confirm with Matt whether legs come on the two removable AVPRO lids. If not, add them as a line item. | Matt (confirm), then Sam | 2026-10-19, with the quote | No | OPEN |

## Cable map

Each row is one signal chain, and panel names match the rear panel. The map is decided except where a route is marked open.

### Video



Rented cameras —SDI→ Panel CAM 4-7 —SDI jumper→ ATEM IN 4-7

ProPresenter Mac —USB-C→ UltraStudio (placement open) —SDI→ ATEM IN 8

ATEM AUX —SDI→ HyperDeck HD Mini —USB-C→ SSD

ATEM OUT 8 —SDI→ Resi —Cat→ AX21 > internet

ATEM multiview —HDMI→ LEFT lid

ProPresenter Mac —HDMI→ LED wall transcoder —Cat x2→ Panel LED 1-2 —field→ LED wall

ProPresenter Mac —USB-C→ Anker hub —HDMI→ Panel CONF OUT —HDMI field→ Confidence monitor

ProPresenter Mac —second screen (route open)→ RIGHT lid

## Audio and Dante

Stage box (DL251) —AES50→ Wing

Wing Dante card —Cat→ Wing case switch (TL-SG1005P, 5-port) —trunk 15ft→ DANTE TRUNK —Cat→ Bay A satellite —Cat + PoE→ AVIO (PoE)

AVIO —XLR pigtails→ ATEM Ch 1-2

ProPresenter Mac —USB-C Cat→ Bay A satellite

ProTools Mac (in case) —Cat (Dante)→ Wing case switch

Bianca's talkback mic —XLR→ Panel TALKBACK —XLR 15ft→ Wing In 8

Wing Out 8 —XLR 15ft→ Panel IEM —XLR→ Bianca's IEM

Wing —USB-C→ ProTools Mac (in case)

## Production network and internet

Venue / Starlink —Cat field→ Panel INTERNET IN —Cat→ AX21 WAN

AX21 LAN —Cat→ TL-SG105 —Cat→ ATEM, HD Mini, streaming laptop, Anker hub

AX21 LAN —Cat→ Resi

AX21 LAN —Cat→ PROD TRUNK —trunk 15ft→ Wing CONTROL

AX21 LAN —Cat→ Panel SPARE

## Power

Wall outlet —5-15P→ Drop 1 (125ft 12/3 SOOW) —5-15R→ UPS CP1500PFCRM2U —mains→ CPS1215RM —5-15P > C13→ Bay A + desk gear, lids

CPS1215RM —feed cord→ POWER > WING outlet —powerCON 15ft 14/3→ Wing

Wall outlet (2nd circuit preferred, not required) —5-15P→ Drop 2 (125ft 12/3 SOOW) —5-15R→ CPPDU511 (Bay B front) —chained→ RS1215-RA (rear) —mains→ Camera chargers

CPPDU511 —extension (details open, not costed)→ Quad box —mains→ Separate charging station

CPS1215RM (battery-backed) —mains→ Laptops